

AI Inventory List

PURPOSE

Summary of all internal AI models and systems deployed or in development at Dëkkal.

Deployed Systems

System	Type	Version	Purpose	Status
Flood Risk Scoring Engine	XGBoost Classifier	v1.0	Address-level flood risk scoring for IARD insurers	Production
Geocoding Service	Rule-Based + Nominatim	v1.0	Convert address strings to geographic coordinates	Production
Feature Extraction Pipeline	GEE Processing Pipeline	v1.0	Extract 11 geospatial features per address from satellite and climate sources	Production
Zone Risk Lookup	Lookup Table	v1.0	Contextual zone-level risk baseline derived from XGBoost training	Production

Systems In Development

System	Type	Version	Purpose	Status
LSTM Temporal Layer	LSTM Neural Network	v0.1	Capture temporal rainfall dynamics for dynamic rainy season scoring	In Development — v2.0
RAG Recommendation Layer	RAG Pipeline (LangChain + Qdrant)	v0.0	Natural language explanations and pricing recommendations	Planned — v2.5

Data Sources

Source	Provider	License	Usage
Sentinel-1 SAR	ESA / Copernicus	CC BY SA (Open)	Flood detection labels
CHIRPS Daily	UCSB / USAID	Open	Precipitation P95/P99
SRTM 30m	NASA / USGS	Open	Terrain elevation, slope
ERA5-Land	ECMWF / Copernicus	Copernicus Licence	Antecedent soil moisture
SoilGrids	ISRIC	CC BY 4.0	Clay fraction, impermeability

Source	Provider	License	Usage
GPM IMERG	NASA EarthData	Open	Extreme rainfall P99 ensemble
Global Flood Database	Google / DFO	CC BY 4.0	Historical flood labels
OpenStreetMap	OSM Contributors	ODbL	Drainage network proximity

Responsible Officer

Babacar Ndao, CTO — responsible for all AI systems listed above.